

ACE uses causal mechanisms to choose experiments

Structural causal model (SCM)



Inputs

- Observed responses
- Known graph and legal actions
- Experiment budget

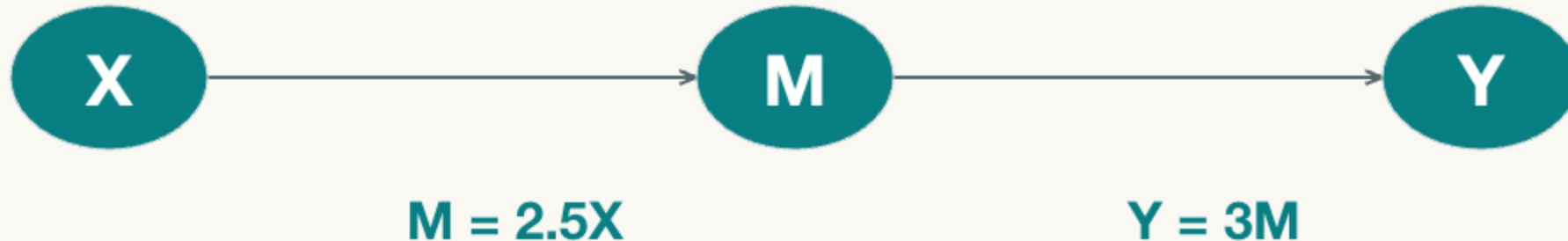
Outputs

- The next intervention
- Updated mechanism models
- Predictions for new actions

Each SCM equation describes one mechanism

$$X_i = f_i(Pa_i, U_i)$$

Parents Pa_i supply inputs. U_i represents outside influences.



Worked example: known graph, unknown slopes, zero disturbances and dimensionless values.

An intervention replaces one mechanism

Ordinary system



Experiment: do(M = 1)



Incoming edge removed

Y still responds to M

One paid response can train several natural mechanisms. It is still one paid response.

Exhaustive grids multiply the work

Illustration: 10 inputs $\times$ 5 settings per input

Naive joint response table

9,765,625

joint input configurations

SCM mechanism tables

225

local parent configurations
9 mechanisms $\times$ 25 each

Representation-size illustration · known graph · five-valued variables · accessible local parent settings

Exact inputs to the three-node demonstration

Mechanism	Member slopes	Mean prediction
M from X	1, 2, 3	$\hat{M} = 2X$
Y from M	2, 3, 4	$\hat{Y} = 3M$

Actions: clamp X or M to -1, 0 or +1

References: -1, 0, +1 Noise proxy: 0.05
Budget: one response Ties: first action in the list

Simplified linear ensemble · six candidate actions · one response budget

The score selects $\text{do}(X = -1)$

Intervention	M contribution	Y contribution	Total
$\text{do}(X = -1)$	0.413437	0.436264	0.849701
$\text{do}(X = 0)$	0.000000	0.000000	0.000000
$\text{do}(X = 1)$	0.413437	0.436264	0.849701
$\text{do}(M = -1)$	0.000000	0.413437	0.413437
$\text{do}(M = 0)$	0.000000	0.000000	0.000000
$\text{do}(M = 1)$	0.000000	0.413437	0.413437

Selected prediction: $M = -2$, $\hat{Y} = -6$

Score contribution: $\text{mean Cov}^2 / (\text{Variance} + \text{noise proxy}) \cdot \text{tied actions follow the declared order}$

The response updates eligible mechanisms

Observed row: $X = -1$, $M = -2.5$, $Y = -7.5$

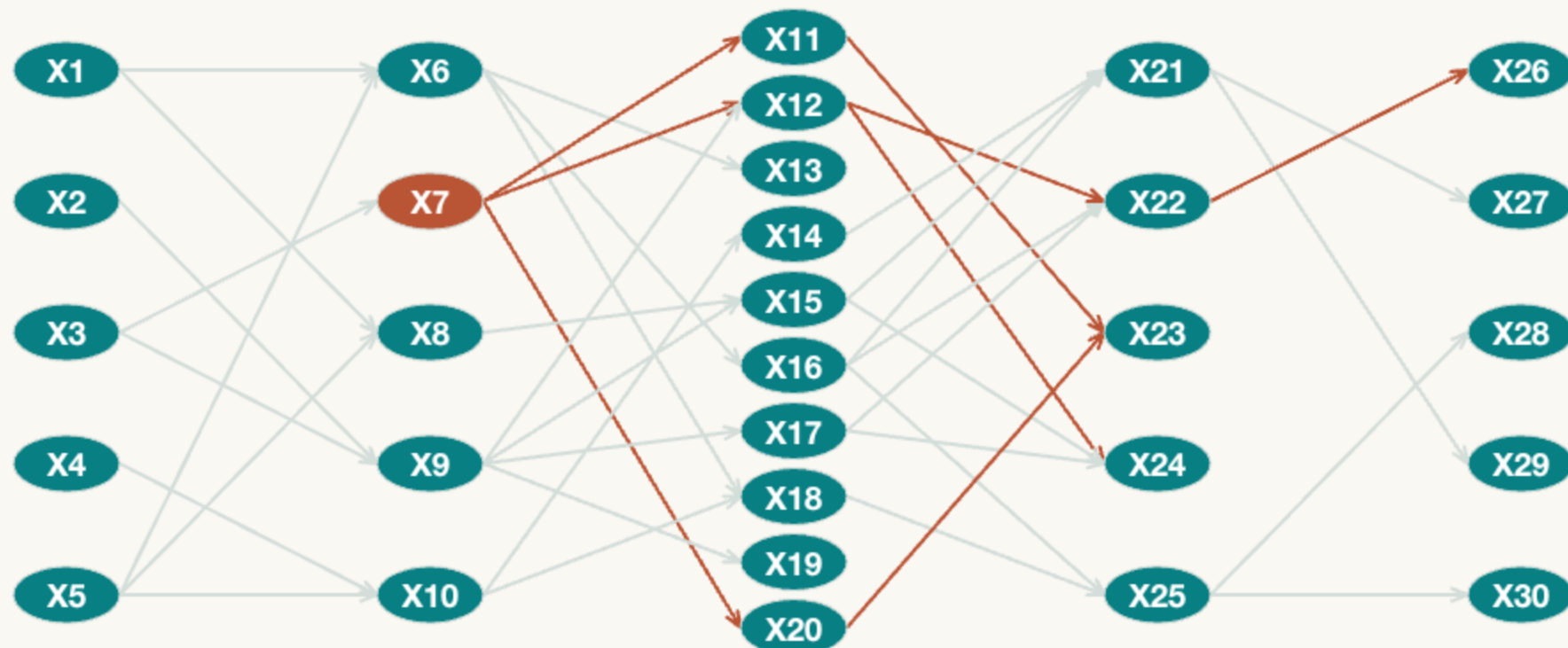
Natural mechanism	Training input	Training label
M	Measured $X = -1$	$M = -2.5$
Y	Measured $M = -2.5$	$Y = -7.5$

After one illustrative gradient update

$M = -2.25$ **$\hat{Y} = -6.75$** **Budget left: 0**

Illustrative linear update · both natural mechanisms learn from the same paid response

A real scenario: 30 linked mechanisms



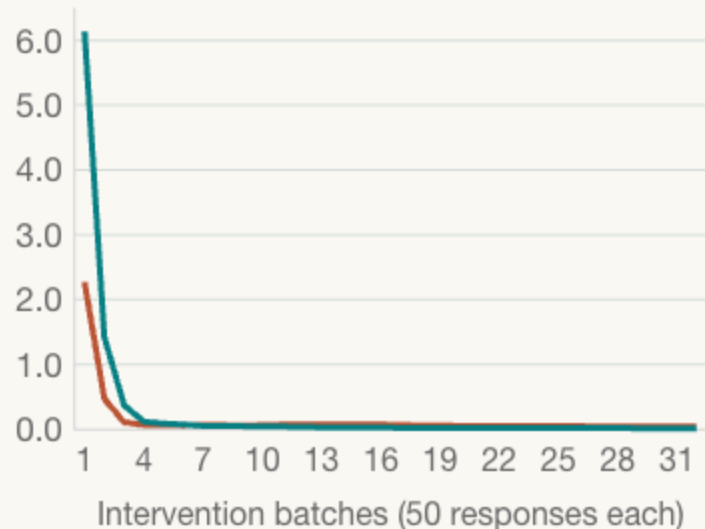
First selected action: $\text{do}(X7 = 3.8571)$ $\rightarrow$ 50 responses

Seed 5000 graph. Coral marks the intervention and outgoing descendant paths; the full study uses 20 systems.

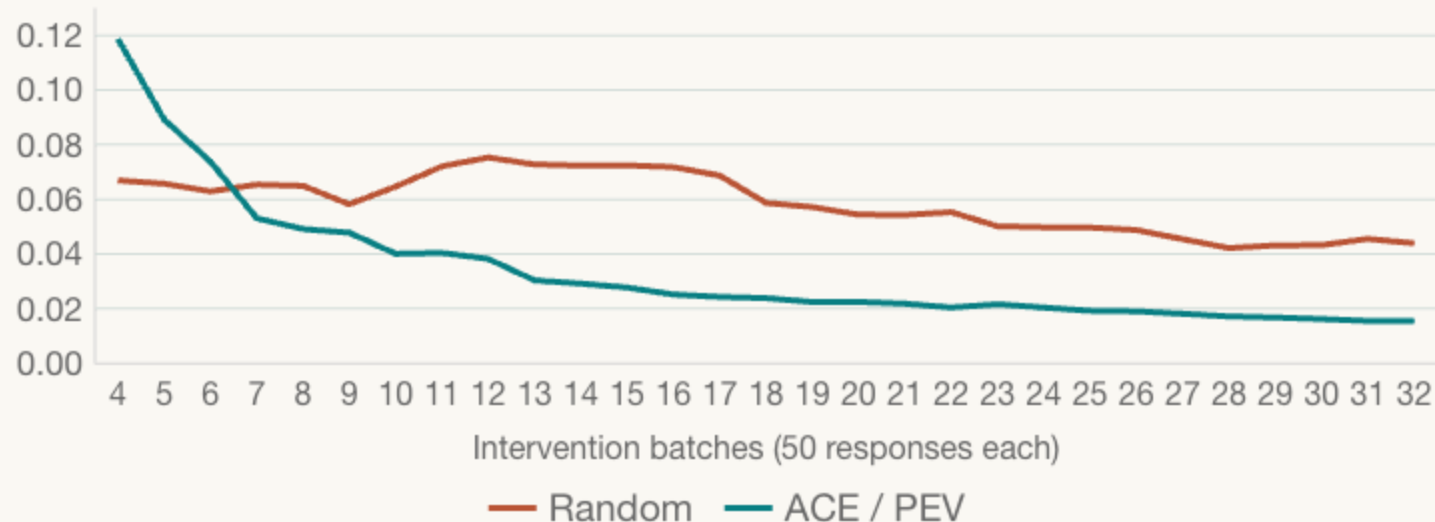
65% lower prediction error than random

Mean mechanism MSE · 20 systems · lower is better

Full recorded curve



Detail: batches 4–32

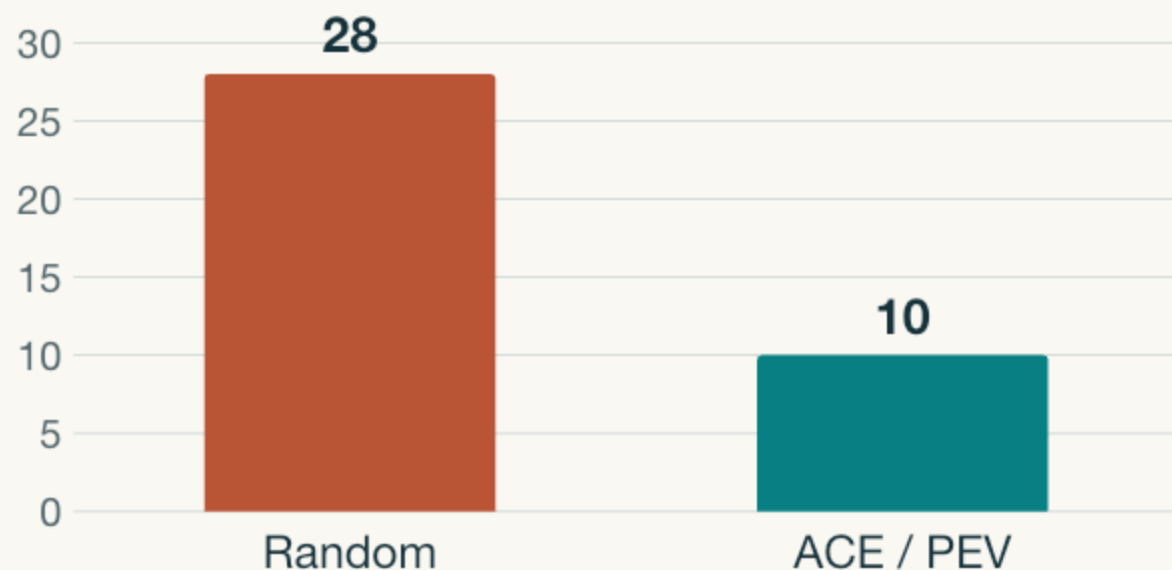


Final MSE: 0.01543 ACE vs 0.04390 random | 19 / 20 wins

20 synthetic causal systems · equal 2,000-response budgets · ACE / PEV versus non-leaf random selection

64% fewer interventions at matched error

First recorded crossing of $\text{MSE} \leq 0.04390$



64.3% fewer

**1,400 → 500
intervention responses**

Including observations:
1,760 → 620 total responses

Intervention batches · 50 responses each

Retrospective comparison of mean learning curves across 20 synthetic systems · target: random's final mean error